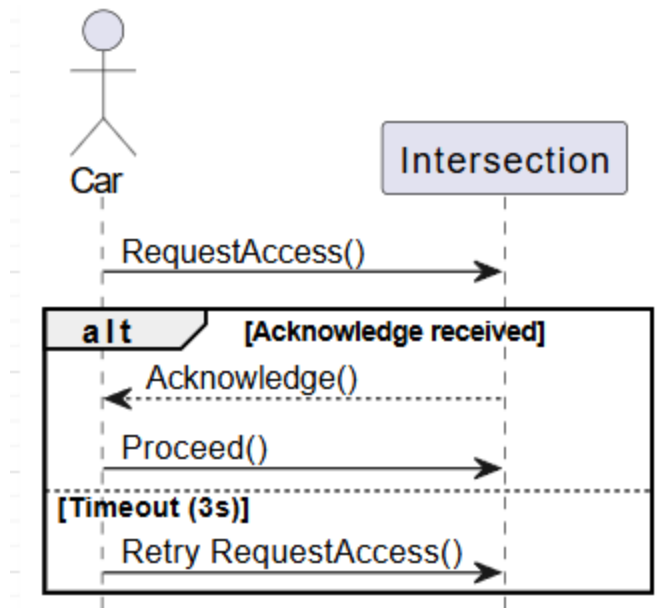




Communication Between Car and Intersection –

The car asks to enter the intersection by sending **RequestAccess()**.

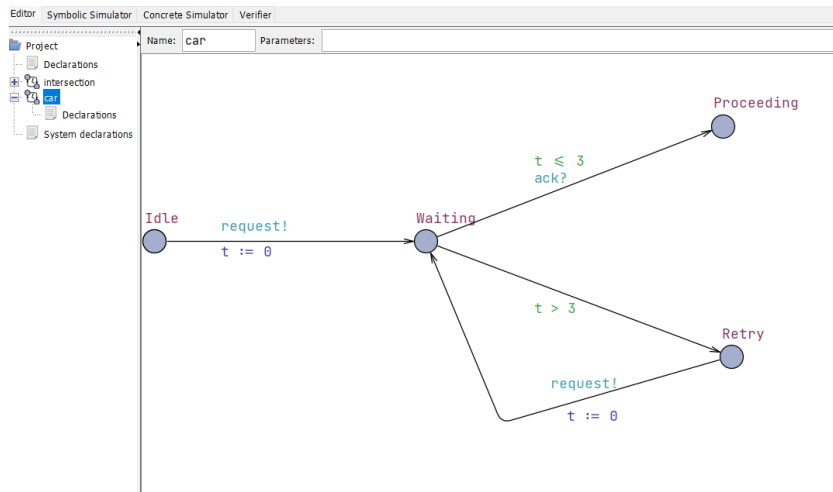
Then two things can happen:

a. If the intersection replies:

- It sends an **Acknowledge()** to say "OK".
- Car sends **Proceed()** and enters.

b. If no reply after 3 seconds:

- The car sends **Retry RequestAccess()** and tries again.



Uppaal

This is a model of a Car waiting for permission.

Idle

- The car is doing nothing.
- It sends a `request!` signal.
- At the same time, the clock `t` is set to 0.

Waiting

- The car is waiting for a reply.
- Two things can happen:

If `t ≤ 3` and it gets `ack?`

- The car goes to Proceeding.
- It got a reply in time — it can move.

If `t > 3`

- It waited too long.

- It goes to Retry and sends request! again.
- Clock t is reset to 0 again.
- It tries waiting one more time.